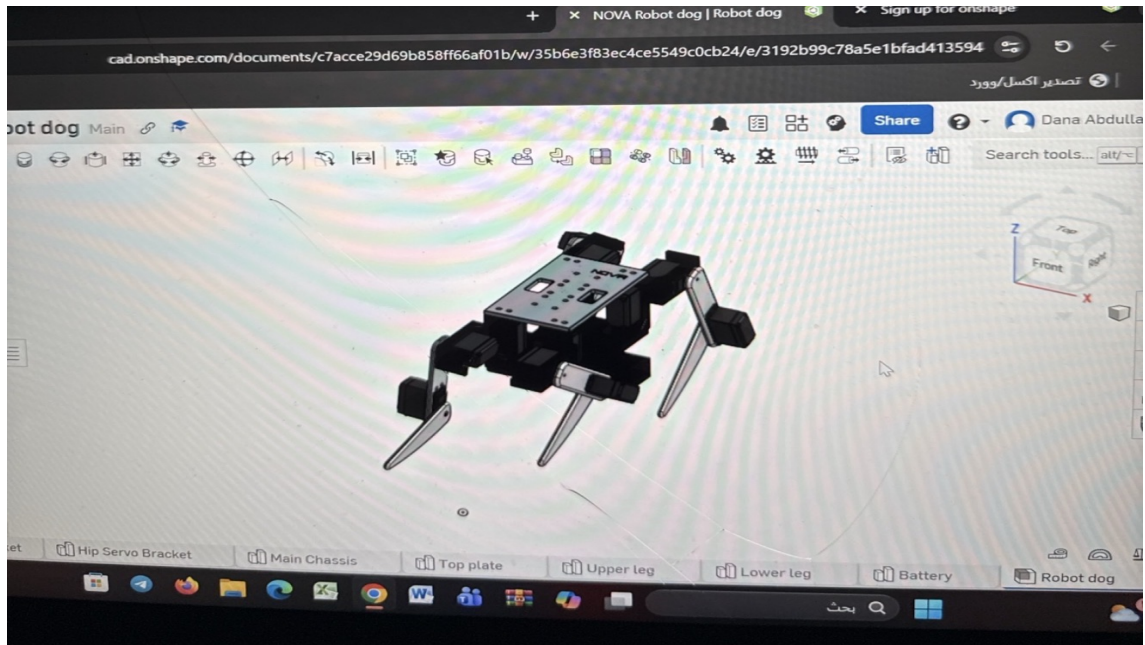


NOVA Robot Dog Project Report



Project Idea

Design a simple quadruped robot dog to demonstrate basic mechanical design concepts and stable walking.

Body and Chassis

A rectangular lightweight chassis with four articulated legs provides structural rigidity and space for electronics.

Leg Design

Each leg consists of an upper and lower segment connected by rotational joints for simple walking.

Degrees of Freedom

$2 \text{ DOF per leg} \times 4 \text{ legs} = 8 \text{ DOF}$.

Motors

Recommended actuators: standard Servo Motors (e.g. MG996R or equivalent).

Preliminary Torque Calculation

Assume leg mass = 0.2 kg, leg length = 0.10 m. Torque = $m \times g \times r = 0.2 \times 9.81 \times 0.10 = 0.196 \text{ N}\cdot\text{m}$. With a safety factor, choose a servo capable of at least 0.5 N·m.

Stability and Center of Gravity

The battery and main components are positioned near the center of the chassis to improve balance and stability.

Walking Method

Alternating diagonal gait (trot gait) is proposed because it is simple and stable.

Expected Mechanical Issues

Joint wear, servo overheating, limited battery life, and reduced stability on uneven surfaces.

Project Link

<https://cad.onshape.com/documents/c7acce29d69b858ff66af01b/w/35b6e3f83ec4ce5549c0cb24/e/3192b99c78a5e1bfad413594>